



**Killara**  
HIGH SCHOOL

STUDENT NUMBER

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# MATHEMATICS

## *Extension 1*

### ASSESSMENT TASK 1

Date: 06/12/2011

**TIME ALLOWED:** *Reading time: 5 minutes*  
*Working time: 60 minutes*

**INSTRUCTIONS:** **SECTION A:** 4 Multiple Choice Questions: Circle your correct choice

**SECTION B:** ANSWER all 3 questions  
Start each question in a new booklet  
Marks may not be given for untidy and poorly arranged work.  
All relevant working should be shown.

***THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM***

**OUTCOMES ASSESSED:**

- 16.1 Graphs of simple polynomials.
- 16.2 The remainder and factor theorems.
- 16.3 The roots and coefficients of a polynomial equation.
- 16.4 Iterative methods for numerical estimation of the roots of a polynomial equation
- 7.4 Mathematical induction. Applications
- 2.8 Simple angle properties of a circle.
- 2.9 Derivation of further angle, chord and tangent results.
- 2.10 Applications of 2.2, 2.3, 2.7, 2.8 and 2.9 to numerical and theoretical problems requiring one or more steps of reasoning.

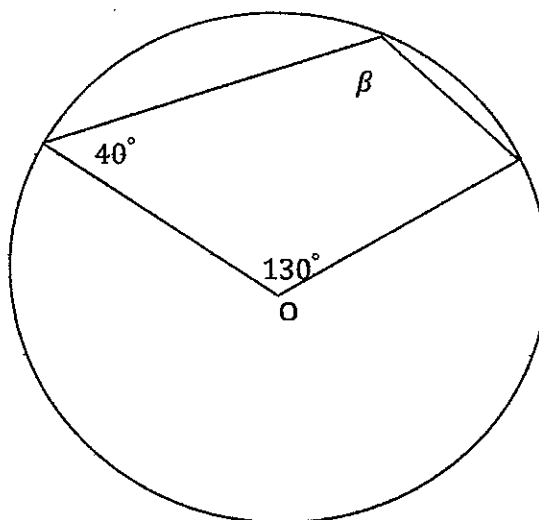
|                                            | MARKS |
|--------------------------------------------|-------|
| <b>SECTION A</b> Multiple Choice questions | /5    |
| <b>SECTION B</b>                           | /15   |
| QUESTION 1 Polynomials                     | /10   |
| QUESTION 2 Circle Geometry                 | /5    |
| QUESTION 3 Mathematical Induction          | /35   |

*This assessment task constitutes 10% of the Higher School Certificate Course Assessment.*

**SECTION I****Marks**

1)

1



O is the centre of the circle. The value of  $\beta$  is:

- A)  $115^\circ$
- B)  $140^\circ$
- C)  $65^\circ$
- D)  $50^\circ$

2)

If the roots of the equation  $x^3 - 8 = 0$  are  $\alpha$ ,  $\beta$  and  $\gamma$ , then the value of

1

$\alpha^3 + \beta^3 + \gamma^3$  is

- A) 8
- B) 6
- C) 24
- D)  $\frac{8}{3}$

Marks  
1

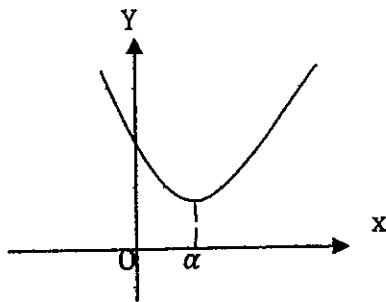
3) Which of the following are true about a cubic function:

- I It must have at least one real zero
- II If it has only one real zero then it must be monotonic
- III If it has a triple root then it must be monotonic

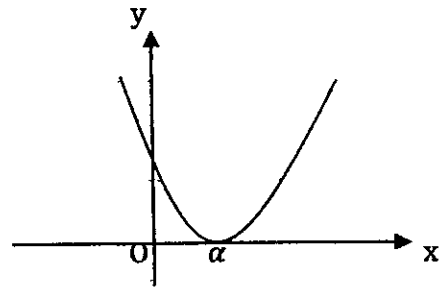
- A) I and II only
- B) I and III only
- C) II and III only
- D) All I, II and III

4) A polynomial  $f(x)$  has a multiple zero at  $x = \alpha$ .

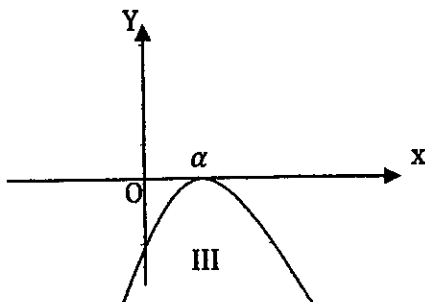
1

Which of the following is a possible graph of  $y = f(x)$ ?

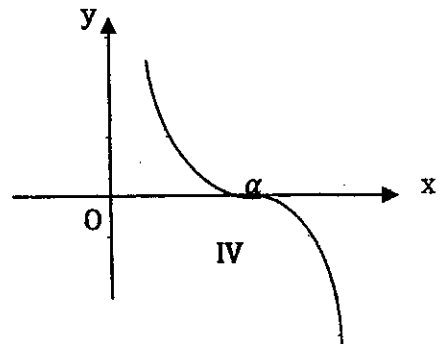
I



II



III



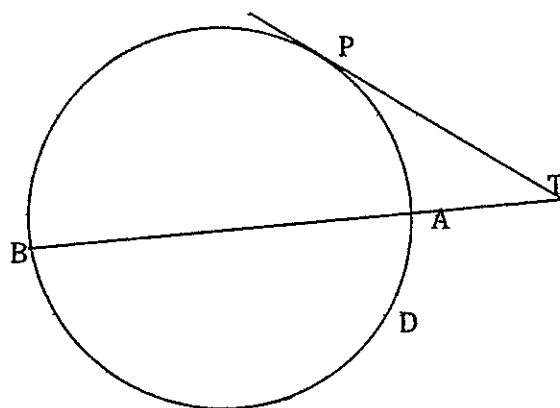
IV

- A) I or II
- B) II or III
- C) III or IV
- D) II or III or IV

Marks

5)

1



TP is a tangent to the circle at P. TAB is a straight line passing through the centre of the circle. If  $PT = 8$  cm and  $TA = 4$  cm, then the radius of the circle is:

- A) 10 cm
- B) 9 cm
- C) 6 cm
- D) 12 cm

**End Of Section I**

## SECTION B

Question 1 (Use a separate writing booklet)

Marks

a) A polynomial  $P(x)$  is given by  $P(x) \equiv x^3 + 3x^2 - 2x + 1$ .

i). Find the remainder when  $P(x)$  is divided by  $(x + 2)(x - 3)$ .

2

ii). Hence, find the values of  $p$  and  $q$ , if the polynomial

$f(x) \equiv x^3 + 3x^2 + px + q$  is exactly divisible by  $(x + 2)(x - 3)$ .

2

b) The roots of the equation  $3x^3 - 16x + 9 = 0$  are  $\alpha, \beta$  and  $\gamma$ .

Find the values of:

i).  $\alpha\beta\gamma$

1

ii).  $\alpha\beta + \beta\gamma + \gamma\alpha$

1

iii).  $\frac{2}{\alpha+\beta} + \frac{2}{\beta+\gamma} + \frac{2}{\gamma+\alpha}$

2

c)

i).  $\alpha$  is a zero of  $f(x) = 3x^3 - 16x + 9$ . Show that  $1 < \alpha < 2$ .

2

ii). Find the turning point of  $y = f(x)$  between 1 and 2.

2

iii). David decided to apply Newton's method to find an approximation for  $\alpha$ . He chose 1.2 as his first approximation.

3

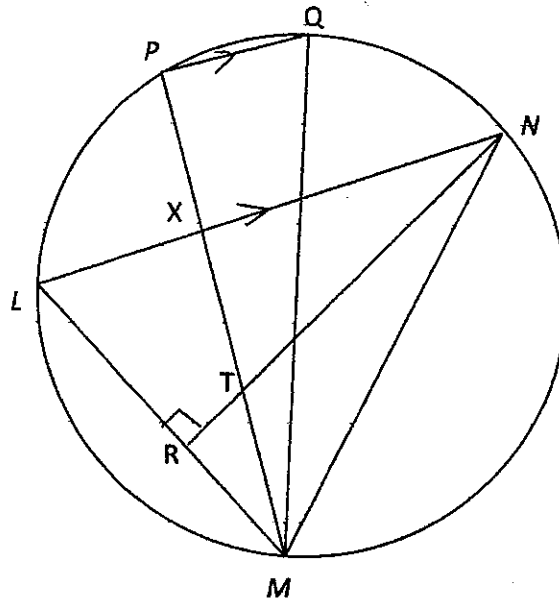
Explain to David with the aid of diagrams why the method would fail.

What range of values should David choose as his first approximation so that he will get a better second approximation?

## Question 2 (Use a separate writing booklet)

Marks

- a)  $MQ$  is a diameter of the circle drawn below.  $PQ \parallel LN$ .  $PM$  cuts  $LN$  at  $X$ .  $NR \perp LM$  and cuts  $PM$  at  $T$ .

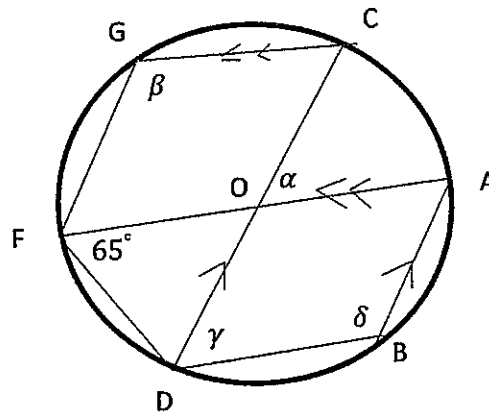


Copy or trace the diagram into your booklet.

- |       |                                                        |   |
|-------|--------------------------------------------------------|---|
| i).   | Explain why $LXTR$ is a cyclic quadrilateral.          | 2 |
| ii).  | Prove, giving reasons, that: $\angle NMQ = \angle PML$ | 3 |
| iii). | Deduce that $PL = QN$                                  | 1 |

Marks

b)



4

O is the centre of the circle.  $DC \parallel BA$  and  $AF \parallel CG$ .  $\angle OFD = 65^\circ$  find the values, giving reasons, of the angles denoted by the pronumerals  $\alpha, \beta, \gamma$  and  $\delta$

Question 3 (Use a separate writing booklet)

i). Prove by mathematical induction that

4

$$\sum_{r=0}^n \frac{1}{(r+1)(r+3)} = \frac{3}{4} - \frac{1}{2(n+2)} - \frac{1}{2(n+3)}$$

for all integers  $n \geq 0$ .

ii). Deduce the value of  $\sum_{r=0}^{\infty} \frac{1}{(r+1)(r+3)}$

1

**End of Paper**

## Section II

Question 1 a) (i)  $(x+2)(x-3) = x^2 - x - 6$

(a) Method 1

$$\begin{array}{r|l} x+4 & x^3 + 3x^2 - 2x + 1 \\ & \underline{x^3 - x^2 - 6x} \\ & 4x^2 + 4x + 1 \\ & \underline{4x^2 - 4x - 24} \\ & 8x + 25 \end{array}$$

$\therefore$  The remainder =  $8x + 25$  ✓

Method 2

Divisor ~~is~~  $(x+2)(x-3)$  is of degree 2

$\therefore$  The remainder is linear (degree 1) of the form  $ax + b$

Let the quotient be  $Q(x)$ .

Writing the division transformation

$$P(x) = x^3 + 3x^2 - 2x + 1 = (x+2)(x-3) \cdot Q(x) + ax + b$$

$$\therefore P(-2) = (-2)^3 + 3(-2)^2 - 2(-2) + 1 = 0 + -2a + b$$

$$\Rightarrow -2a + b = 9 \quad \text{--- (1)}$$

$$P(3) = (3)^3 + 3(3)^2 - 2(3) + 1 = 0 + 3a + b$$

$$\Rightarrow 3a + b = 49 \quad \text{--- (2)}$$

Solving equations (1) and (2) simultaneously,

$$a = 8, b = 25$$

$\therefore$  The remainder =  $8x + 25$

Q<sub>1</sub> (i) (ii)

$$x^3 + 3x^2 - 2x + 1 = (x+2)(x-3) \cdot Q(x) + 8x + 25$$

$$\Rightarrow x^3 + 3x^2 - 10x - 24 = (x+2)(x-3) \cdot Q(x)$$

(i.e)  $x^3 + 3x^2 - 10x - 24$  is divisible by  $(x+2)(x-3)$   
 Since  $x^3 + 3x^2 + px + q$  is divisible by  $(x+2)(x-3)$

$$x^3 + 3x^2 + px + q = x^3 + 3x^2 - 10x - 24$$

matching the coefficients of corresponding terms,

$$\underline{p = -10} \text{ and } \underline{q = -24}$$

(b)  $3x^3 - 16x + 9 = 0$  roots  $\alpha, \beta$  and  $\gamma$

(i)  $\alpha\beta\gamma = -9/3 = -3 // \checkmark$

(ii)  $\alpha\beta + \beta\gamma + \gamma\alpha = -16/3 // \checkmark$

(iii)  $\frac{2}{\alpha+\beta} + \frac{2}{\beta+\gamma} + \frac{2}{\gamma+\alpha}$

Since  $\alpha+\beta+\gamma = 0$   $\alpha+\beta = -\gamma$ ,  $\beta+\gamma = -\alpha$  and  $\gamma+\alpha = -\beta$

$$\Rightarrow -\frac{2}{\gamma} - \frac{2}{\alpha} - \frac{2}{\beta} \checkmark$$

$$= -\frac{2(\alpha\beta + \beta\gamma + \gamma\alpha)}{\alpha\beta\gamma}$$

$$= \frac{-2(-16/3)}{-3}$$

$$= \underline{\underline{-32/9}} \checkmark$$

Q. (c) i)  $f(x) = 3x^3 - 16x + 9$

$f(1) = 3 - 16 + 9 = -4$

$f(2) = 24 - 32 + 9 = 1$

$f(x)$  is a cubic function.

$f(x)$  is continuous in the interval  $1 \leq x \leq 2$

Since  $f(1) < 0$  and  $f(2) > 0$ , there exists a such that  $1 < x < 2$  and  $f(x) = 0$

(ie)  $1 < x < 2$ , and  $x$  is a root.

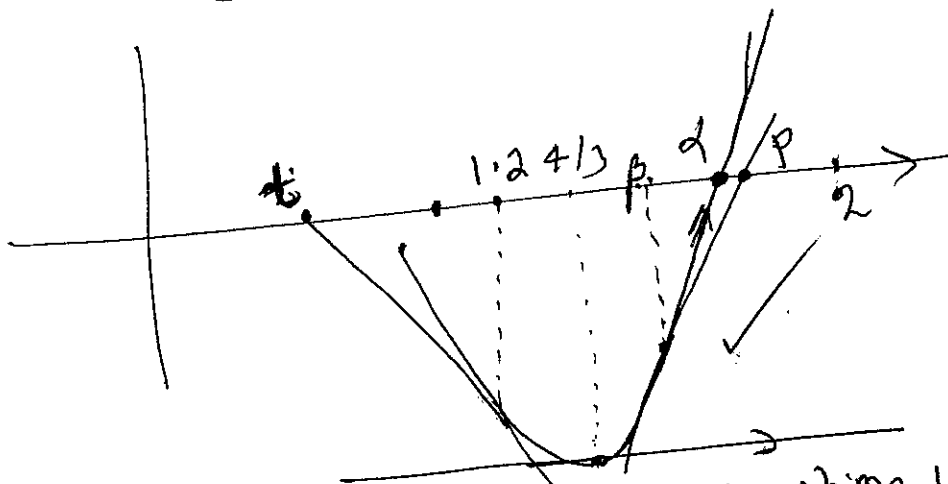
ii)  $f'(x) = 9x^2 - 16$

$f'(x) = 0 \Rightarrow x = \frac{4}{3}$  or  $x = -\frac{4}{3}$

$f''(x) = 18x$

at  $x = \frac{4}{3}$   $f''(x) = 18 \times \frac{4}{3} > 0$   $\therefore (\frac{4}{3}, -\frac{47}{9})$  is a minimum turning point

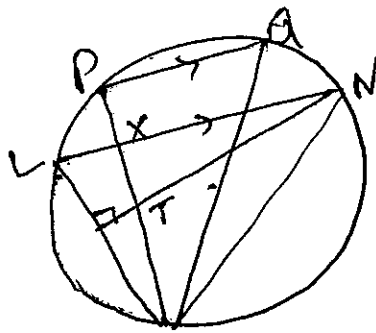
(iii) Below is a rough sketch of  ~~$y = f(x)$~~  the graph of  $y = f(x)$  close to the interval  $[1, 2]$ .



Newton's method of approximation is based on finding the  $x$ -intercept of the tangent drawn to the curve at the first approximation.

- $\therefore$  If the first approximation = 1.2 then the second approximation is  $x = t$  which is further away from 1.2
- If the first approximation =  $\frac{4}{3}$  then the tangent at  $x = \frac{4}{3}$  will not have an  $x$ -intercept  $\therefore$  second approximation cannot be found
- If the first approximation is between  $\frac{4}{3}$  and 2 then the second approximation is  $p$  which is a better approximation than the first at  $x = p$ .

Section II  
Question 2(a)



- (i)  $\angle QPM = 90^\circ$  (angle subtended by diameter ~~MA~~  
MQ on the circle)  
 $\angle QPM = \angle PXL$  (alternate angles on || lines,  $PQ \parallel LM$ )  
 $\therefore \angle PXL = 90^\circ$   
 $\angle TRL = 90^\circ$  (given)

(ii)  $\angle PXL = \angle TRL$  ✓  
 $\therefore LXTR$  is a cyclic quadrilateral as the  
exterior angle  $PXL =$  interior opposite angle  $TRL$

- (ii)  $\angle NMQ = \angle QLP$  (angles subtended on the circle by  
the same arc  $\widehat{NQ}$ )  
But  $\angle QLP = \angle PQL$  (alternate angles on || lines;  $PQ \parallel LN$ )  
and  $\angle PQL = \angle PML$  (angles subtended on the circle  
by the same arc  $\widehat{LP}$ )  
 $\Rightarrow \angle NMQ = \angle PML$

- (iii)  $\angle NMQ$  is the angle subtended on the circle by the  
chord  $QN$   
 $\angle PML$  is the angle subtended on the circle by the  
chord  $PL$

Since  $\angle NMQ = \angle PML$  (Proven in (ii))  
 $PL = QN$  (angles subtended on the circle  
by equal chords are equal)

Q2 (b)

In  $\triangle OFD$

$OF = OD$  (radii of same circle)

$$\therefore \angle OFD = \angle ODF = 65^\circ$$

$$\text{and } \angle FOD = 180^\circ - 2 \times 65^\circ$$

$$= 50^\circ$$

$\angle FOD = x$  vertically opposite angle

$$\therefore \boxed{x = 50^\circ}$$

$$\text{Reflex } \angle FOC = 180^\circ + x$$

$$= 230^\circ$$

$$\beta = \frac{1}{2} \times \text{reflex } \angle FOC$$

$$= \frac{1}{2} \times 230^\circ$$

$$\boxed{\beta = 115^\circ}$$

angle subtended by arc  $FDBAC$  on the circle is half the angle subtended at the centre by same arc

$FDBA$  is a cyclic quadrilateral

$$\therefore \delta + 65^\circ = 180^\circ \quad (\text{opp. } \angle \text{ s of a cyclic quad are supplementary})$$

$$\boxed{\delta = 115^\circ}$$

$$\gamma + \delta = 180^\circ \quad (\text{co-interior angles on } \parallel \text{ lines } DO \parallel BA \text{ given})$$

$$\therefore \gamma = 180^\circ - 115^\circ$$

$$\boxed{\gamma = 65^\circ}$$

### Question 3

$$(1) \sum_{r=0}^n \frac{1}{(r+1)(r+3)} = \frac{3}{4} - \frac{1}{2(n+2)} - \frac{1}{2(n+3)}$$

For  $n=0$

$$\text{LHS} = \frac{1}{1 \cdot 3} = \frac{1}{3}$$
$$\text{RHS} = \frac{3}{4} - \frac{1}{4} - \frac{1}{6}$$
$$= \frac{1}{3}$$

$$\therefore \text{LHS} = \text{RHS}$$

$\therefore$  True for  $n=0$

Assume true for  $n=k$

$$\text{i.e. } \sum_{r=0}^k \frac{1}{(r+1)(r+3)} = \frac{3}{4} - \frac{1}{2(k+2)} - \frac{1}{2(k+3)}$$

for  $n=k+1$

$$\begin{aligned} \sum_{r=0}^{k+1} \frac{1}{(r+1)(r+3)} &= \sum_{r=0}^k \frac{1}{(r+1)(r+3)} + \frac{1}{(k+2)(k+4)} \\ &= \frac{3}{4} - \frac{1}{2(k+2)} - \frac{1}{2(k+3)} + \frac{1}{(k+2)(k+4)} \\ &= \frac{3}{4} - \frac{1}{2(k+3)} + \frac{1}{(k+2)(k+4)} - \frac{1}{2(k+2)} \\ &= \frac{3}{4} - \frac{1}{2(k+3)} + \frac{2 - (k+4)}{2(k+2)(k+4)} \\ &= \frac{3}{4} - \frac{1}{2(k+3)} - \frac{(k+2)}{2(k+2)(k+4)} \\ &= \frac{3}{4} - \frac{1}{2(k+3)} - \frac{1}{2(k+4)} \end{aligned}$$

$$\therefore \text{True} = \frac{3}{4} - \frac{1}{2[(k+1)+2]} - \frac{1}{2[(k+1)+3]}$$

$\therefore$  True for  $n=k+1$

Concluding statement

Q3  
(ii)

$$\sum_{r=0}^{\infty} \frac{1}{(r+1)(r+2)} = \lim_{n \rightarrow \infty} \frac{3}{4} - \frac{1}{2(n+2)} - \frac{1}{2(n+3)}$$

$$\text{as } n \rightarrow \infty, \frac{1}{2(n+2)} \rightarrow 0$$

$$\text{and } \frac{1}{2(n+3)} \rightarrow 0$$

$$\therefore \sum_{r=0}^{\infty} \frac{1}{(r+1)(r+2)} = \underline{\underline{\frac{3}{4}}} \checkmark$$